MODULE 55 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

# Practice Exam: Domain 3

Fifteen original questions on cost-optimised architecture. Pause, decide, then watch.

DOMAIN 3 of 5 EXAM WEIGHT 15% SET 3 of 6

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# How to use this set

- **Allow 2.4 minutes per question**: 36 minutes for the set.
- **Write your answer down** before you unpause.
- **The cheapest option must still meet every requirement.**
- The last slide maps every question to the module that teaches it.

> ON THE EXAM
>
> A cost answer that breaks a requirement is wrong, however cheap it is.

A ticketing company runs 60 instances all year. For a three-day sale it needs 200 more instances of one type in one Availability Zone, and launches must not fail.

It wants the lowest cost for the year-round fleet.

**What should the solutions architect recommend?**

A. A Compute Savings Plan sized for 260 instances

B. Spot Instances for the sale

C. Reserved Instances for 260 instances

D. An On-Demand Capacity Reservation for the sale, and a Savings Plan covering the year-round 60

◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

A Savings Plan for 260
*No capacity guarantee; pays for 200 idle all year.*

B Spot for the sale
*Can be interrupted or unavailable.*

C RIs for 260
*Commits a year for a three-day need.*

D Reservation plus Savings Plan
Guaranteed launches; discount on the steady base.

THE SIGNAL

A Savings Plan never guarantees capacity. Reserve for the event, commit for the baseline. Module 25.

A company runs EC2 instances across several families, plus Fargate tasks and Lambda functions. Over the next three years it expects to change instance families and possibly move workloads to another Region.

Spend is steady at about 40 dollars an hour.

## Which commitment gives the most flexibility?

A EC2 Instance Savings Plans

B Compute Savings Plans

C Standard Reserved Instances

D Convertible Reserved Instances

◆ PAUSE AND DECIDE

# Which commitment gives the most flexibility?

A EC2 Instance SP
*Locked to a family in a Region.*

B Compute Savings Plans
Any family, size, Region, plus Fargate and Lambda.

C Standard RIs
*Fixed attributes; EC2 only.*

D Convertible RIs
*Exchangeable, but EC2 only.*

THE SIGNAL

Changing families, Regions, or using Fargate and Lambda: Compute Savings Plans. Module 25.

> A company spends steadily on Amazon RDS for PostgreSQL and Aurora MySQL. Next year it plans to move some databases between engines and instance families.
>
> It wants a commitment discount that survives those moves.

**What should the solutions architect recommend?**

A Database Savings Plans

B RDS Reserved Instances for each current instance

C Compute Savings Plans

D Aurora Serverless for everything

◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

**A** Database Savings Plans
Applies across engines and instance families.

**B** RDS Reserved Instances
*Tied to engine, family and Region.*

**C** Compute Savings Plans
*Do not apply to database services.*

**D** Move to serverless
*Not a commitment discount; may cost more.*

THE SIGNAL

Database spend that will change engines or families: Database Savings Plans. Module 25.

A research group runs nightly genome analysis jobs that checkpoint every few minutes and can restart. Jobs can run on many instance types.

The group wants the lowest compute cost with few interruptions.

**What should the solutions architect recommend?**

A On-Demand Instances of one type

B Reserved Instances for the nightly peak

C Spot Instances across many instance types, with the price-capacity-optimized allocation strategy

D Spot Instances of a single instance type

◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

A On-Demand
*Pays full price for interruptible work.*

B Reserved
*Pays all day for a nightly job.*

C Diversified Spot
Deepest discount; diversity reduces interruptions.

D Single-type Spot
*One pool: more interruptions.*

THE SIGNAL

Fault-tolerant, flexible work: Spot, diversified across instance types. Module 26.

> A payments company's Lambda functions run at a steady, very high volume, all day, every day. Its finance team already holds Savings Plans and wants EC2-style pricing for this compute.
>
> The team wants to keep the Lambda programming model.

**What should the solutions architect recommend?**

A Rewrite the functions as containers on EC2

B Lambda Managed Instances for these functions

C Provisioned concurrency on every function

D Increase memory to shorten duration

◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

A Rewrite for EC2
*Gives up the Lambda model.*

B Lambda Managed Instances
Lambda model on EC2 capacity and pricing.

C Provisioned concurrency
*Reduces cold starts; still Lambda pricing.*

D More memory
*May help a little; not the pricing change.*

THE SIGNAL

Steady, very high-volume functions that want EC2 pricing and commitments: Lambda Managed Instances. Module 26.

A legal research assistant on Amazon Bedrock sends a 40-page practice guide as context with every question. The guide changes monthly, and users ask thousands of questions a day.

Input token costs are the largest line on the bill.

**What should the solutions architect do first?**

A Switch to a larger model

B Fine-tune a model on the guide

C Use batch inference for every question

D Enable prompt caching for the repeated guide

◆ PAUSE AND DECIDE

## What should the solutions architect do first?

A Larger model

*Costs more per token.*

B Fine-tune

*Training is out of scope and the guide changes.*

C Batch inference

*Users need answers now.*

D Prompt caching

Cached prefix tokens are billed at a lower rate.

THE SIGNAL

The same long prefix in every request: prompt caching. Module 27.

A support chatbot on Amazon Bedrock gets a mix of trivial questions and complex troubleshooting. All requests currently go to the most capable model in a family.

The team wants lower cost without training anything or losing quality on hard questions.

**What should the solutions architect use?**

A Intelligent prompt routing across models in the family

B Model distillation

C Send everything to the smallest model

D Provisioned Throughput for the large model

◆ PAUSE AND DECIDE

## What should the solutions architect use?

**A** Prompt routing
Routes each request to the model that suits it.

**B** Distillation
*Requires a training job; suits one narrow task.*

**C** Smallest model
*Hard questions lose quality.*

**D** Provisioned Throughput
*Steady capacity, not a cheaper mix.*

THE SIGNAL

Mixed difficulty, no training: routing. One narrow task at huge volume: distillation. Module 27.

> Applications in private subnets read and write 300 TB a month to S3 in the same Region. The bill shows large NAT gateway data processing charges.
>
> Security wants traffic to stay on the AWS network.

**What should the solutions architect do?**

A Move the applications to public subnets

B Add a second NAT gateway

C Add an S3 gateway endpoint to the private subnets' route tables

D Use S3 Transfer Acceleration

◆ PAUSE AND DECIDE

## What should the solutions architect do?

A Public subnets

*Weakens security.*

B Second NAT gateway

*Doubles the cost.*

C S3 gateway endpoint

No processing charge; stays on AWS.

D Transfer Acceleration

*For long-distance uploads; adds cost.*

THE SIGNAL

S3 traffic through NAT gateways: a gateway endpoint, which has no charge. Module 28.

A streaming service delivers video worldwide through CloudFront with long TTLs already set. The origin is still overloaded by cache misses arriving from many edge locations at once when new episodes launch.

The team wants to reduce origin load and cost.

**What should the solutions architect enable?**

A Shorter TTLs

B A second origin in another Region

C Lambda@Edge to rewrite cache keys

D CloudFront Origin Shield in the Region closest to the origin

◆ PAUSE AND DECIDE

## What should the solutions architect enable?

A Shorter TTLs

*More misses, not fewer.*

B Second origin

*Doubles origin cost.*

C Lambda@Edge

*Helps only if cache keys are fragmented.*

D Origin Shield

Collapses misses from all locations into one layer.

THE SIGNAL

Caching already tuned, origin still flooded by misses from many locations: Origin Shield. Module 28.

A company replicates an entire 2 PB bucket to a second Region for disaster recovery. Half the data is temporary processing output under the tmp/ prefix, deleted within a week.

Replication and storage costs in the second Region are higher than expected.

**What should the solutions architect change?**

A Switch to Same-Region Replication

B Scope the replication rule to exclude temporary data, for example by prefix or tag

C Replicate with S3 Batch Operations once a month

D Turn off versioning in the destination

◆ PAUSE AND DECIDE

## What should the solutions architect change?

A Same-Region Replication
*No longer protects against a Regional failure.*

B Filtered replication rule
Stops paying to replicate data nobody needs.

C Monthly batch copy
*Loses continuous protection.*

D No versioning
*Replication requires versioning on both buckets.*

THE SIGNAL

Replicate only what recovery needs: filter rules by prefix or tag. Module 29.

> Finance activated a team cost allocation tag today. The tag has existed on resources for a year, but cost reports only show it from today.
>
> Finance needs the previous months broken down by team for a budget review.

**What should the solutions architect do?**

A Request a cost allocation tag backfill for the previous months

B Re-tag every resource

C Wait twelve months for history to build

D Export Cost Explorer data and edit it by hand

◆ PAUSE AND DECIDE

## What should the solutions architect do?

**A** Tag backfill
Applies the tag to up to 12 months of past data.

**B** Re-tag
*Tags already exist; reports still miss history.*

**C** Wait a year
*Too late for the review.*

**D** Edit by hand
*Error-prone and unauditable.*

THE SIGNAL

A tag existed but was activated late: backfill cost allocation tags. Module 29.

A company's AWS spend varies by team and by month. Last quarter, a misconfigured job tripled one team's data transfer costs for two weeks before anyone noticed.

Finance wants to hear about unusual spend quickly, without setting thresholds for every team and service.

**What should the solutions architect configure?**

A AWS Budgets with a fixed threshold per team

B A monthly cost report emailed to each team

C CloudWatch billing alarms per service

D AWS Cost Anomaly Detection monitors with alert subscriptions

◆ PAUSE AND DECIDE

## What should the solutions architect configure?

A Budgets per team
*Needs a threshold for each, and spend varies.*

B Monthly report
*Weeks too late.*

C Billing alarms
*Fixed thresholds again.*

D Cost Anomaly Detection
Learns normal patterns and alerts on unusual spend.

THE SIGNAL

A known limit: Budgets. A surprise nobody set a threshold for: Cost Anomaly Detection. Module 30.

> Twelve teams share one EKS cluster with GPU and CPU node groups. Finance will charge each team for its share of the cluster.
>
> Teams must accept the allocation as fair.

**What should the solutions architect implement?**

A Split the bill evenly across the teams

B Tag each node group by team

C Enable split cost allocation data for EKS, allocating node costs by each pod's resource requests and use

D Give each team its own cluster

◆ PAUSE AND DECIDE

## What should the solutions architect implement?

A Even split

*Not fair; teams will dispute it.*

B Tag node groups

*Nodes are shared, so tags cannot divide them.*

C Split cost allocation data

Measured per-pod allocation, including GPUs.

D Cluster per team

*Costly duplication to solve a reporting problem.*

THE SIGNAL

Chargeback needs measured allocation. Shared clusters: split cost allocation data. Module 30.

A media archive holds 400 TB in S3 Standard. Some objects are read daily, most are never read, and access patterns change unpredictably when old stories trend.

Retrieval must stay instant, with no retrieval fees.

**Which storage class should the solutions architect choose?**

A S3 Glacier Deep Archive

B S3 Intelligent-Tiering

C S3 Standard-IA for everything

D S3 One Zone-IA

◆ PAUSE AND DECIDE

## Which storage class should the solutions architect choose?

A **Deep Archive**
*Hours to retrieve.*

B **Intelligent-Tiering**
Moves objects by access, no retrieval fees.

C **Standard-IA**
*Retrieval fees on popular objects.*

D **One Zone-IA**
*Less durable, still retrieval fees.*

THE SIGNAL

Unknown or changing access, instant retrieval, no retrieval fees: Intelligent-Tiering. Module 31.

> A platform team ingests 20 TB a month of verbose debug logs into CloudWatch Logs. They are queried with Logs Insights perhaps twice a month during incidents.
>
> The team does not need metric filters or subscription filters on these logs.

**What should the solutions architect change?**

A Use the CloudWatch Logs Infrequent Access log class for these log groups

B Delete the logs after one day

C Send the logs to an EC2-hosted Elasticsearch cluster

D Reduce the retention period to seven days

◆ PAUSE AND DECIDE

## What should the solutions architect change?

**A** Infrequent Access class
Lower ingestion price, Logs Insights still works.

**B** Delete after a day
*Loses incident data.*

**C** Self-hosted cluster
*Adds servers to run.*

**D** Shorter retention
*Ingestion, not storage, is the main cost.*

THE SIGNAL

Rarely queried logs without filters: the Infrequent Access log class. Module 31.

# Where to revisit

| IF YOU MISSED | REWATCH |
|---|---|
| Q1, Q2, Q3 | Module 25: Pricing models and commitments |
| Q4, Q5 | Module 26: Compute cost |
| Q6, Q7 | Module 27: AI cost |
| Q8, Q9 | Module 28: Delivery and data transfer |
| Q10, Q11 | Module 29: Tagging and replication cost |
| Q12, Q13 | Module 30: Visibility and chargeback |
| Q14, Q15 | Module 31: Storage, database and observability cost |

ON THE EXAM

12 or more out of 15 is on track. Domain 3 is 15%, but cost clues appear in every domain.